DIP BHATTACHARJEE

CSE Undergraduate | Full-Stack & AI Developer | Data Analysis | ML Research

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github.com/dipjee678

PROFESSIONAL SUMMARY

Computer Science and Engineering undergraduate (CGPA 3.83, Dean's Award) who builds **full-stack web applications with AI built in**, from a live multilingual AI CRM to a PHP + MySQL e-commerce store, and works across **data analysis**, **AI/ML research** and practical IT systems. Author of three IEEE conference papers in 2026 on medical image classification, ML-enabled IoT monitoring, and Monte Carlo risk estimation, each built on collecting, cleaning and statistically analysing real data in Python. Experienced with SQL and relational databases, Excel reporting and cloud deployment, and comfortable explaining technical work clearly to non-technical people from experience as an instructor and tutor.

EDUCATION

Bachelor of Science in Computer Science and Engineering — **Aug 2023 – Present**
American International University-Bangladesh — *CGPA: 3.83/4.00 · Dean's Award*

Major: Computational Theory · **Relevant Courses:** Machine Learning, Natural Language Processing, Graph Theory, Data Structures and Algorithms, Object-Oriented Programming with Java, Programming in C++, Discrete Mathematics.

HSC, Science — *Mohipal Government College* — *GPA 4.89/5.00 · 2020 – 2022*
SSC, Science — *Batisa Secondary School* — *GPA 4.89/5.00 · 2018 – 2020*

RESEARCH & PUBLICATIONS

Development of a Cloud-Linked Low-Cost Machine Learning-Enabled Weather Monitoring System
Published in the Proceedings of IEEE TENSYMP 2026 · DOI: 10.1109/TENSYMP68909.2026.11704037
Designed a low-cost sensing system that sends readings to the cloud and uses ML to monitor and predict weather conditions, combining hardware, connectivity and analysis of time-series sensor data.

Progressive Two-Stage Transfer Learning with Dual Imbalance Compensation for Interpretable Enamel Caries Classification
Accepted for Presentation at the 5th IEEE International Conference on Signal Processing, Information, Communication and Systems (SPICSCON 2026)
Two-stage fine-tuning with imbalance correction at both data and loss level, plus explainable-AI visualisation so clinicians can see where the model looked.

Diagnostic-Guided Monte Carlo Risk Estimation Using GARCH, EGARCH, and Bootstrap Models
Accepted for Presentation at the 3rd IEEE International Conference on Computing, Applications and Systems (COMPAS 2026)
Statistical diagnostics on financial time-series data select between volatility and bootstrap models before Monte Carlo simulation, giving risk estimates that fit the data.

TECHNICAL SKILLS

Web & Software Development

Frontend:	HTML, CSS, JavaScript, responsive design (hand-built portfolio website)
Backend & APIs:	PHP (MVC), serverless functions (Vercel), REST APIs, session-based authentication
AI Integration:	LLM APIs (Google Gemini), schema-validated outputs (Zod), retries and model fallback

Data & Analytics

Data Analysis:	Data Collection, Cleaning and Preprocessing, Data Quality Checks, Exploratory Data Analysis, Statistical Modelling, Monte Carlo Simulation, Reporting
Python for Data:	pandas, NumPy, Matplotlib, Jupyter Notebook
Databases & SQL:	SQL (SELECT, JOIN, GROUP BY, Subqueries), MySQL, Relational Database Design, CRUD Applications
Spreadsheets:	Microsoft Excel (Formulas, Pivot Tables, Charts)

AI & Machine Learning

Machine Learning:	Deep Learning, Transfer Learning, Classification, Model Training & Evaluation, Class Imbalance Handling, Explainable AI
NLP & LLMs:	Text Preprocessing and Classification, Structured Outputs, Prompt Design

IT & Systems

Deployment:	Git & GitHub, serverless deployment (Vercel), installing and configuring applications
Networking & IoT:	TCP/IP, subnetting, DHCP, DNS, LAN/Wi-Fi troubleshooting; sensor hardware and cloud connectivity
Security Practice:	Server-side handling of API keys and secrets, input validation with schemas, safe fallbacks when services fail
Programming:	Python, SQL, PHP, C, C++, Java, C#, JavaScript
Other Tools:	Microsoft Office (Excel, Word, PowerPoint), AutoCAD, Adobe Photoshop

PROJECTS

LeadDesk: Multilingual AI Enquiry CRM *JavaScript, Serverless API, Gemini, Zod, Vercel*

- Built and deployed a full-stack web CRM, with a JavaScript frontend and a serverless API backend, that turns customer messages (WhatsApp, Facebook, web forms) in Bangla, Banglish or English into structured leads using an LLM whose outputs are validated against Zod schemas.
- Classifies messages (sales, support, spam, job seeker), drafts replies in the customer's language, and tracks follow-ups with a drag-and-drop pipeline, 48-hour reminders and weekly AI summaries.
- Designed for reliability: API key kept on the server, automatic model fallback and retries, and a keyword-based mode that keeps the app working without AI.
- Live demo: biai-crm-eight.vercel.app · Source: github.com/dipjee678/LeadDesk

DeshiDrip: E-commerce Clothing Store *Team project · PHP, MySQL, MVC*

- Full-stack online store with a customer storefront and an admin dashboard (products, orders, customers, pending orders).
- **My role:** designed the 7-table relational MySQL schema (users, categories, products, cart, orders, order items, payments) and built the cart, checkout and order/payment flow, including size selection, order line items, payment records and stock updates on every order. Source: github.com/dipjee678/DeshiDrip-Clothing

Personal Portfolio Website *HTML, CSS, JavaScript*

- Hand-built responsive site with light/dark themes and an in-browser neural network demo written from scratch (JavaScript, Canvas, no libraries). Live: dipjee678.github.io/2nd-Protfolio

Hospital Management System *Java, OOP*

- Developed a system for patient records, appointments, billing and staff management, using object-oriented design for structured, maintainable code.

Poultry Farm Management System *Java, SQL Database*

- Developed a database-driven application with a relational schema, SQL queries and full create, read, update and delete workflows for day-to-day farm operations.

EXPERIENCE

Instructor *NextGen Coaching Center for Admission Preparation*

- Taught core admission-preparation subjects and developed mock tests and quizzes to improve students' speed and accuracy.
- Provided one-on-one support, diagnosing where each student was stuck and explaining difficult topics clearly and patiently.

Freelance Tutor *Programming & Problem Solving*

- Mentored students in Python, JavaScript and C++, helping them set up their tools, debug their code and work through data structures, algorithms and practical projects.
- Designed personalised learning plans based on each student's level and goals.

ADDITIONAL INFORMATION

Languages:	Bangla (Native), English (Fluent), Hindi (Proficient in Listening, Basic Speaking)
Research Interests:	Machine Learning, Data Analytics, NLP, Computer Vision, Explainable AI, Medical Image Analysis, IoT-based Intelligent Systems
Availability:	Open to internships and roster-based schedules